

Problem-Solution Fit

Date	01 July 2026
Team ID	XXXXX
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] Disaster management authorities, meteorological departments, and residents/communities in flood-prone districts.	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] Limited to historical rainfall and weather datasets; no real-time IoT sensor or satellite feed integration.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Manual/conventional forecasting: familiar but slow and often inaccurate. Statistical models: fast but rigid, low precision. Generic weather apps: not flood-specific.
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] Lack of timely, accurate flood early-warning - recurring every monsoon/rainfall season, sometimes with only hours of notice.	9. PROBLEM ROOT / CAUSE [RC] Absence of an automated, data-driven flood risk prediction system integrated into an accessible platform.	7. BEHAVIOR + ITS INTENSITY [BE] Frequent monitoring of rainfall and weather reports; high anxiety and urgency during heavy rainfall/monsoon periods.
3. TRIGGERS TO ACT [TR] Heavy rainfall alerts, rising water levels, and past flood damage or casualties in the region.	10. YOUR SOLUTION [SL] Rising Waters: a Flask-based ML web application using a trained XGBoost model (96.55% accuracy) to instantly predict flood risk from rainfall, cloud visibility, and seasonal rainfall data - enabling evacuation advisories hours in advance.	8. CHANNELS OF BEHAVIOR [CH] ONLINE Web application / disaster management portal, mobile access.
4. EMOTIONS BEFORE / AFTER [EM] Before: anxious, uncertain about safety. After: reassured, better prepared with advance warning.		OFFLINE Local disaster management office, radio/TV/SMS alerts, community announcements.